VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Application ID

VK26-AU-FD-0015

NadiSense: a ring that checks your pulse, oxygen and temperature while you sleep

Frontier Diagnostics: The Next Vital Signs, Team Ring of Care, Rajalakshmi Engineering College

Problem Context & Project Overview

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1 in 4	2 of 3	>85%	~80%
ADULTS HYPERTENSIVE	OF THEM UNDIAGNOSED	SLEEP APNEA UNDIAGNOSED	RURAL SPECIALIST SHORTFALL

Sources: NFHS-5 analyses (medRxiv 2024); ResMed economic model, ISPOR Europe 2025; MoHFW Health Dynamics of India 2022-23.

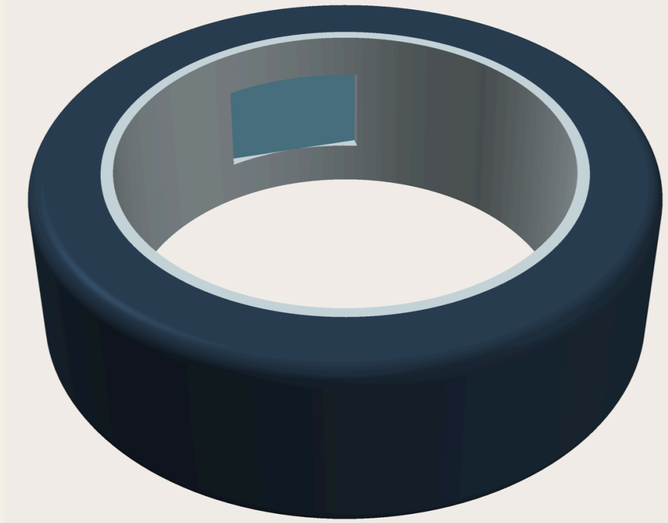


Figure: NadiSense ring v2, CAD render (AI Generated)

Target Beneficiary & Problem Statement

- **Target User:** adults aged 30–60 in semi-urban and rural India, starting with Chennai's peri-urban PHC/CHC areas.
- **Pain Point:** hypertension, irregular heart rhythm and sleep apnea are only caught by one-off tests: a BP cuff, an ECG or an overnight sleep study.
- **Gap:** nobody measures at night, and rural clinics often have no specialist to run these tests.
- **Existing Solutions:** current smart rings are fitness gadgets for city users, with no link to a PHC referral.

Project: NadiSense | Track: Frontier Diagnostics | Core idea: a pulse, SpO_2 and temperature ring worn overnight.

User Journey: Before State

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1 in 11	< 1 in 5	₹19,500 Cr	4,413 / 21,964
HYPERTENSIVES CONTROLLED	RECEIVE TREATMENT	YEARLY COST OF OSA	RURAL CHC SPECIALISTS

Sources: NFHS-5 hypertension cascade (medRxiv); ResMed economic model, ISPOR Europe 2025; MoHFW Health Dynamics of India 2022-23.

Silent signs
Snoring, fatigue, palpitations dismissed as normal

→

No routine check
BP checked rarely; nobody measures at night

→

Facility-bound
Cuff at PHC, ECG/Holter, overnight sleep lab

→

Late detection
First sign is often a stroke or heart event

Why spot checks fail

Paroxysmal AF and drops in night-time oxygen happen when nothing is measuring. Sleep studies are expensive and rare outside cities, so 2 in 3 people with hypertension and over 85% of people with sleep apnea are never diagnosed.

Bottleneck: every step depends on the patient going to a clinic, so problems are found late.

User Journey: After State

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~8 h	0	< 60 s	≈ ₹100
DATA PER NIGHT	CLINIC VISITS TO SCREEN	SYNC TO RISK REPORT	HARDWARE PER SCREENING

Wear the ring
No cuff, no clinic, no straps

→

Passive capture
Pulse wave, HRV, SpO_2, skin temp

→

Risk flags
Stiffness, AF-type rhythm, apnea

→

Early referral
Report for PHC doctor or teleconsult

Quantified Efficiency (design targets)

Metric	Today	NadiSense
Nights monitored / month	0–1	Every night worn
Visits needed to screen	1–3	0 (1 to confirm)
Time to risk report	Days–weeks	< 60 s after sync

Affordable Deployment Model

A PHC or ASHA worker lends one ring out for a week at a time. One ₹4,370 ring can screen about 45 people a year, which is **about ₹100 of hardware per person**. No sleep lab is needed.

Scope: NadiSense flags risk and refers to a doctor; it does not diagnose. We will test these targets in Stage 3.

System Architecture: Block Diagram

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Sensing
MAX86141 with red/IR LEDs and a photodiode, TMP117, LIS2DW12

→

Control & Power
nRF52840, 12 mAh Li-Po and BQ25120A charger

→

Firmware Logic
Continuous at night, short checks by day; skips windows with motion

→

App & Analytics
Syncs over BLE, runs the risk models, makes a doctor's report

Mapping Nadi Pariksha to PPG Features

Traditional parameter	NadiSense measurement	Screening use
Vega (rate)	Heart rate from beat detection	Baseline deviation
Sthiratva / Tala (stability, rhythm)	Beat-to-beat interval variability	AF-type irregular rhythm
Kathinya (arterial hardness)	PPG stiffness index	Hypertension-risk trend
Gati / Bala (movement, force)	Waveform shape, pulse amplitude	Personal baseline
Tapamana (temperature)	Skin temperature (TMP117)	Fever / illness flag

Basis: Nadi Tarangini studies found the radial pulse looks like a finger PPG wave and linked Kathinya to the stiffness index.

System Architecture: Target Specs & Hardware

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≈ 3 g	40 µA	6–10 days	4 signals
RING WEIGHT	AVERAGE CURRENT	BATTERY PER CHARGE	PPG, SpO_2, TEMP, MOTION

Engineering Target Specifications & Subsystems

Parameter	Target Spec	Selected Hardware / Protocol
Optical sensing	25 sps, 19-bit ADC, AFE < 25 µA	MAX86141 (SPI) + red 660 / IR 880 nm LEDs + Si PD
Heart rate	±3 bpm at rest vs reference	On-device beat detection, nRF52840 FPU
SpO_2	±3% (70–100%)	Red/IR ratio, calibrated vs reference oximeter
Skin temperature	±0.1 °C sensor accuracy	TMP117, I²C, one reading per minute
Motion rejection	Gate windows above motion limit	LIS2DW12 accelerometer (~1 µA)
Power & battery	40 µA average, 6–10 days per charge	12 mAh curved Li-Po (245 mm³), BQ25120A
Wireless	One batched sync per night	BLE 5.0, custom GATT service
Size & weight	ID 19 mm, OD 26 mm, 8 mm wide; ≈ 3 g	SLA resin shell, polyimide flex PCB

Why MAX86141: it draws 11 µA where the MAX30101 drew 600 µA, and it needs no 5 V rail for a green LED.

CAD Design & Mechanical Specs

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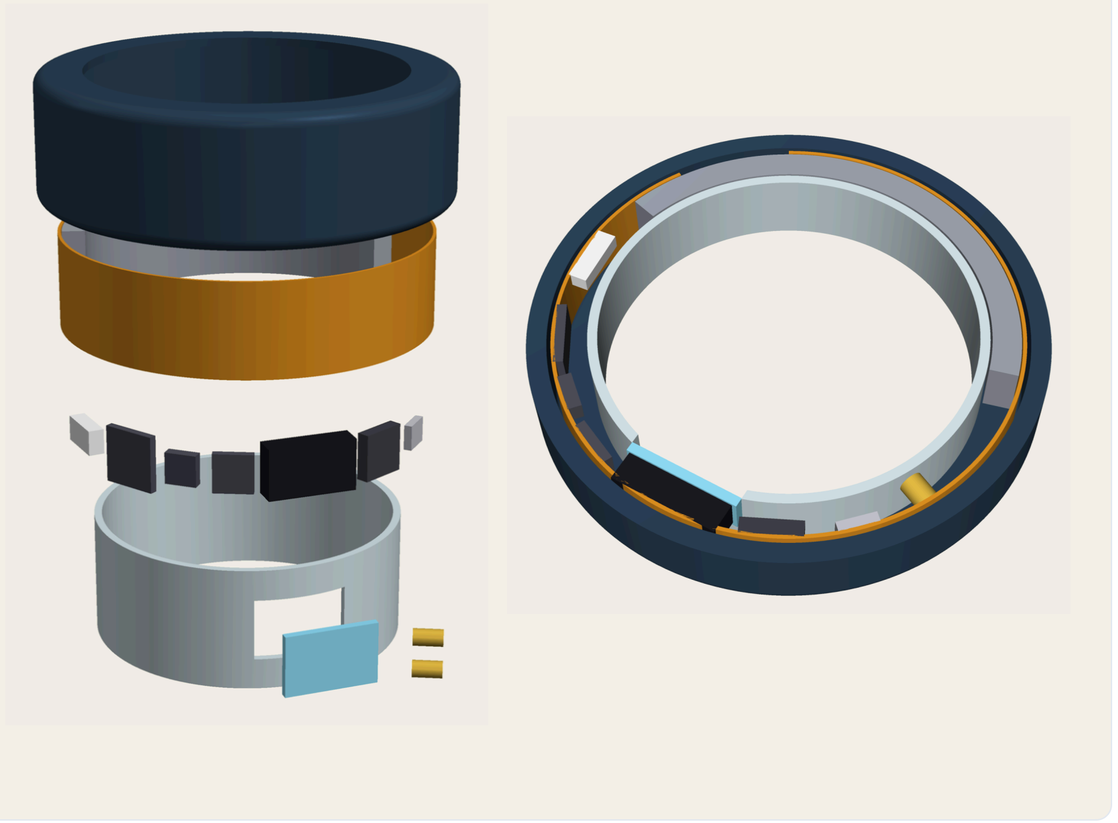


Figure: exploded assembly and half-section, ring v2 (AI Generated)

Mechanical Design & Material Selection

- **Envelope:** ID 19 mm, OD 26 mm, 8 mm wide; wall = 0.5 liner + 2.0 cavity + 1.0 shell.
- **Shell:** biocompatible SLA resin U-channel, 0.8 mm outer and 0.3 mm inner fillets.
- **Liner:** clear resin with a 6 × 4 mm optical window sealed in clear epoxy.
- **Layout:** 245 mm³ curved Li-Po on the back of the finger; optics on the palm side; flex PCB wraps 330°.
- **Mass:** ≈ 3 g estimated from CAD volumes.

Design Iteration: we first planned a 20–40 mAh cell, but the CAD cavity holds only 245 mm³, so we resized it to 12 mAh.

CAD Design: FEA Stress Analysis

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5.9	8.5 MPa	0.12 mm	0.8%
FACTOR OF SAFETY	PEAK VON MISES	MAX DEFLECTION	MESH CONVERGENCE

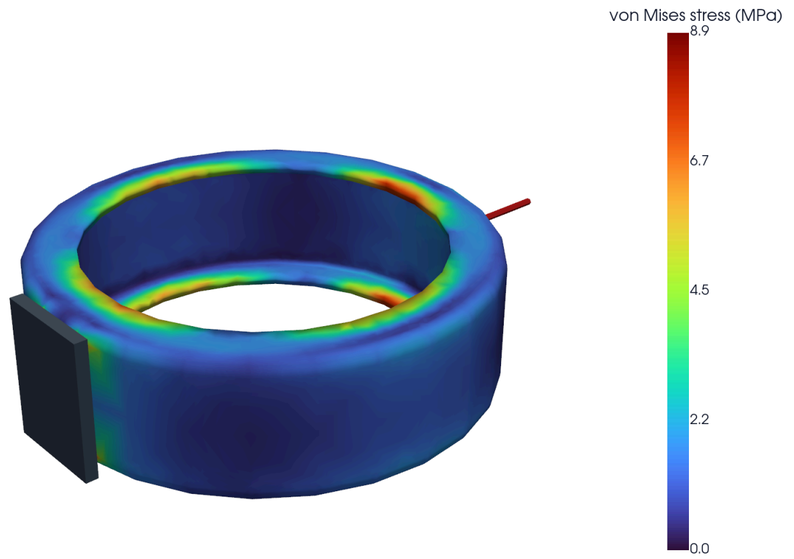


Figure: von Mises stress, 50 N squeeze, 90k-element mesh (AI Generated)

Mechanical Validation & FEA Report

- **Load:** 50 N squeeze across the ring; 6 mm support patch opposite.
- **Material:** rigid SLA resin, E = 2.5 GPa, UTS 50 MPa (assumed); shell only, conservative.

Mesh	Elements	DOFs	Peak σ
Coarse	22,945	135,009	9.35 MPa
Medium	61,197	329,184	8.53 MPa
Fine	90,507	457,659	8.46 MPa

Design Iteration: we added 0.3 mm fillets at the inner channel corners before the analysis to remove sharp stress points.

Electronics Schematics & Power Budget

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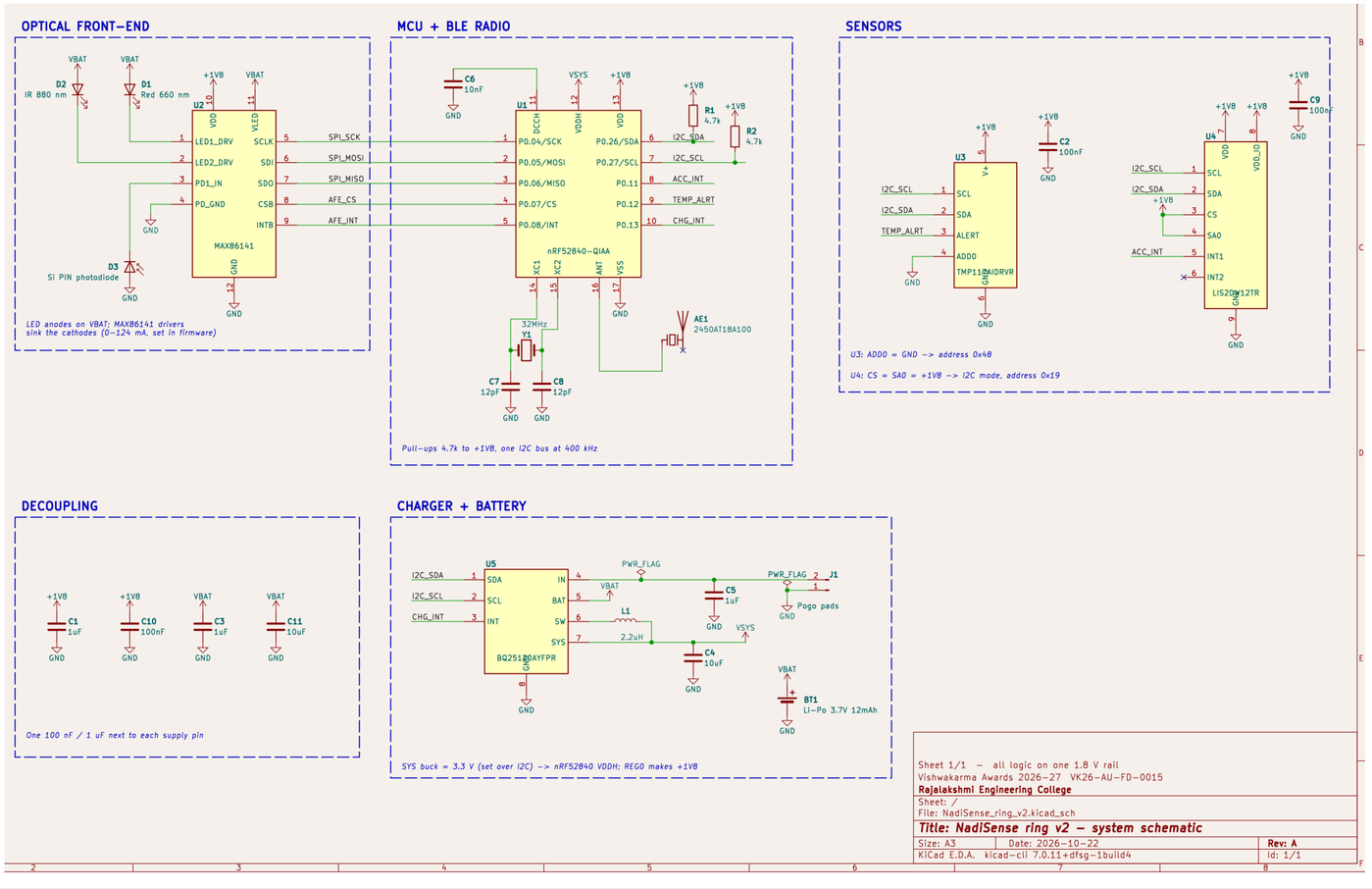


Figure: KiCad 7 schematic sheet, Rev A, ERC clean (AI Generated)

Power Budget

Mode	Avg current
Night, 8 h (continuous)	98 µA
Day, 16 h (spot checks)	12 µA
24 h average	**40 µA**
Life, 12 mAh × 80%	9.9 days
Worst case*	6.0 days

Night current breakdown (98 µA)



■ LEDs 60 µA ■ AFE 22 µA
■ MCU 10 µA ■ Other 7 µA

*20 mA LEDs, 2× CPU time

Power Distribution: the BQ25120A buck gives 3.3 V to the nRF52840, whose regulator makes 1.8 V for all logic. LEDs run off the battery.

Electronics BOM & Procurement

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Itemized Bill of Materials (one ring)

Ref	Component	Part / Specification	Sourcing / Lead Time	Est. (INR)
U1	BLE SoC	nRF52840, WLCSP	Distributor (5–7 days)	₹420
U2	Optical AFE	MAX86141, WLP	Distributor (5–7 days)	₹520
U3	Temperature	TMP117, ±0.1 °C	Distributor (5–7 days)	₹260
U4	Accelerometer	LIS2DW12	Distributor (5–7 days)	₹140
U5	Charger	BQ25120A	Distributor (5–7 days)	₹180
D1–D3	Optics	Red/IR SMD LEDs, Si photodiode	Distributor (5–7 days)	₹220
Y1, AE1, L1	RF + passives	32 MHz crystal, chip antenna, 2.2 µH, R/C	Distributor (5–7 days)	₹230
PCB	Flex PCB	2-layer polyimide, 0.2 mm	Imported (10–14 days)	₹700
BT1	Battery	12 mAh curved Li-Po (custom)	Imported (3–4 weeks)	₹1,200
MECH	Shell, liner, potting	SLA resin, epoxy, silicone, pogo pads	Local (4 days)	₹500
	Total per unit			**₹4,370**

Procurement Risk: the custom curved cell has the longest lead time, so it is ordered first, in Month 1.

Your Maker Journey

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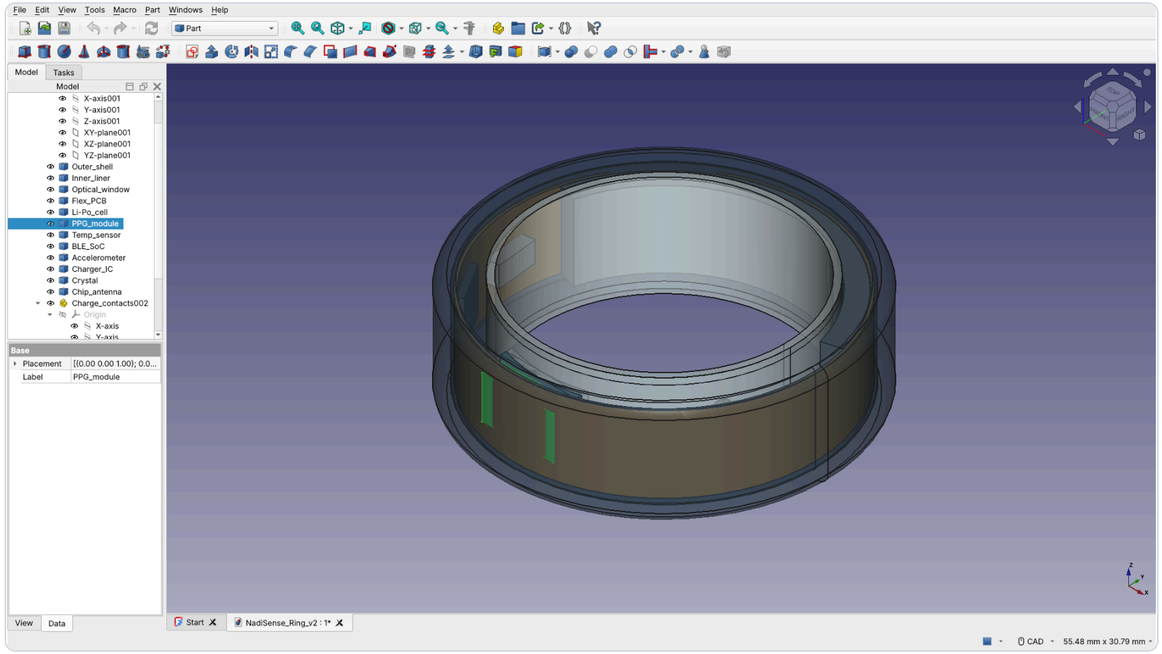
Stage 1 to Stage 2 Design Iterations

- **Form factor:** moved from a handheld reader to a ring so we can record through the night.
- **Sensing:** added SpO_2, skin temperature and motion sensing to the PPG.
- **Optical AFE:** the power budget ruled out the MAX30101 (600 µA), so we moved to the MAX86141 (11 µA).
- **Battery:** cut from 20–40 mAh to 12 mAh to fit the CAD cavity; still 6–10 days per charge.
- **Shell:** 0.3 mm internal fillets added; FEA safety factor 5.9 at 50 N.
- **Software:** companion app live at nadisense-app.vercel.app in 8 Indian languages, running on simulated sensor data until the ring is built.
- **Repository:** github.com/yeswantsakthi-hub/NadiSense, with STEP files, KiCad project, FEA scripts, power budget and app code.

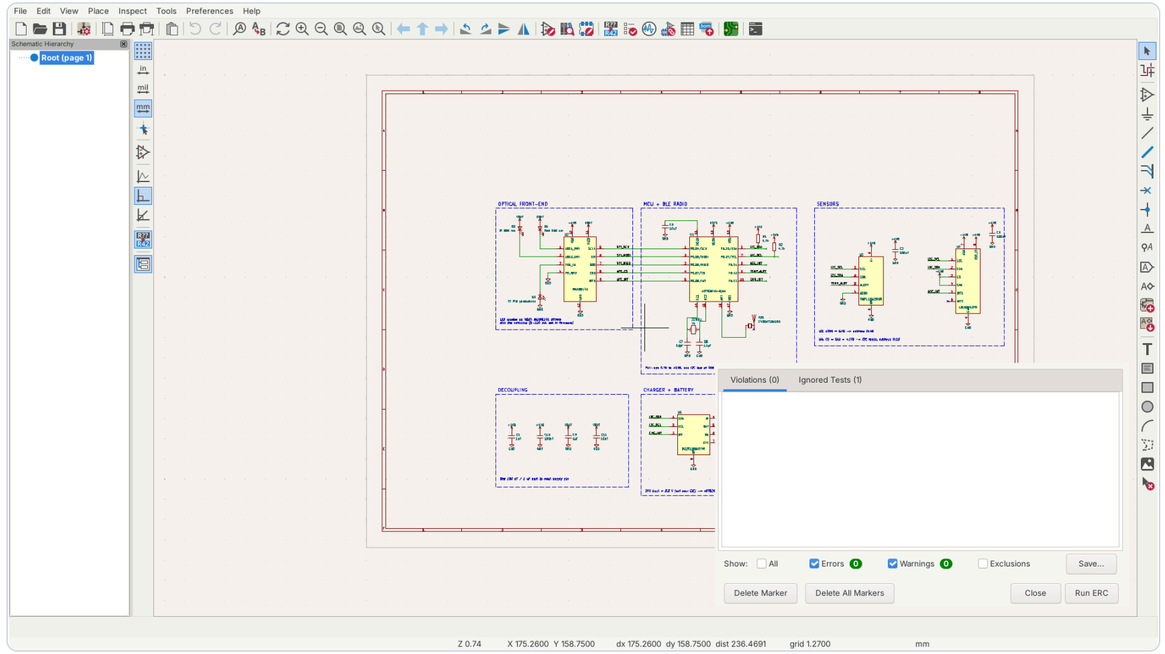
Institutional Sign-Off: design reviewed by Faculty Guide [Name], Dept. of [ ], Rajalakshmi Engineering College, Chennai.

Your Maker Journey: Design Tools

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FreeCAD 1.0: ring v2 STEP assembly, 13 parts in the model tree (AI Generated)



KiCad 7: Electrical Rules Check, 0 errors, 0 warnings (AI Generated)

CAD: parametric model exported to STEP and checked in FreeCAD; same file used for the FEA.

ERC: first run found 6 errors and 14 warnings; we added power flags, fixed pin types and a project library.

Files: STEP, KiCad project, ERC report and netlist are in github.com/yeswantsakthi-hub/NadiSense.

Safety & Hazard Mitigation

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Design FMEA (Severity × Occurrence × Detection, each 1–10)

Failure Mode	Potential Hazard	Embedded Fail-Safe / Mitigation	S	O	D	RPN
Cell overheat	Burn at the finger	Charger thermal limit; cut-off above 43 °C	9	2	3	54
LED skin heating	Warmth, discomfort	LED current cap; duty-cycled sampling	5	2	4	40
Motion artifact	False alert	IMU gating; alert only on repeat episodes	4	6	4	96
Missed event	False reassurance	Never shows “all clear”; clinical check	8	4	6	**192**
Data exposure	Privacy breach	On-phone processing; encrypted BLE and storage	6	2	5	60

Physical Fail-safes

Fully potted electronics, rounded edges, no exposed conductors except gold charging pads; ship mode on the charger.

Standards Compliance

IEC 60601-1-11, ISO 80601-2-61, ISO 10993, IEC 62133-2, IEC 62304, CDSCO MDR 2017, DPDP Act 2023

Highest RPN: missed event (192). The app never says “all clear”, and every flag ends in a referral.

Roadmap: 4-Month Fabrication Plan

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	Nov 2026	Dec 2026	Jan 2027	Feb 2027
Procurement	Custom Li-Po, ICs, flex PCB order			
Mechanical	Print shells, fit trials, drop test			
Electronics	MAX86141 eval kit, flex PCB layout, ring v2 build			
Firmware & App		Algorithms on public PPG data, BLE sync, reports		
Validation			Bench tests, then overnight pilot	
Milestone				**Stage 3 demo ready**

Stage 3 Target: a working ring streaming live HR, SpO_2 and temperature, producing a screening report from one night's wear.

Technical Challenges & Risks

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Motion Artifacts

Risk: finger movement corrupts the PPG signal far more than wrist placement; daytime readings may be unusable.

SpO_2 Skin-Tone Bias

Risk: pulse oximeters are documented to overestimate SpO_2 in darker-skinned patients, which matters for Indian users.

Power vs Sensing

Risk: the cavity fits only a 12 mAh cell; the first AFE choice (600 µA) would drain it in under a day.

No Labelled Nadi Data

Risk: no public dataset links PPG to traditional pulse readings, and PPG alone cannot measure absolute blood pressure.

Verification Plan: every risk is tested on the bench against a reference oximeter and BP cuff in January 2027.

Potential Solutions & Risk Mitigation

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Motion Artifacts

Mitigation: IMU-gated windows; screen mainly during sleep, when the hand is still.

Fallback: IR-only HR in daytime; tighter fit with a sizing kit.

SpO_2 Skin-Tone Bias

Mitigation: calibrate against a reference oximeter across varied skin tones.

Fallback: flag apnea from relative desaturation drops, not absolute SpO_2.

Battery Life

Mitigation: MAX86141 at 25 sps and one BLE sync per night; 6–10 days by our power budget.

Fallback: night-only mode; 2.5 mm cavity for a ~16 mAh cell.

Data Gap

Mitigation: train on public datasets (PPG-DaLiA, PPG-BP, MIMIC waveforms, MESA).

Fallback: rule-based HRV and desaturation scores; collect our own labelled pilot data.

Scope: NadiSense tracks arterial-stiffness *trends* and refers for a cuff reading; it never reports a BP number.